SCI-AST v0.1

Engineering conformance view

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Use of this view. This is a requirements-oriented view of the same six canonical modules as the scientist-facing rationale. It defines prospective conformance obligations and falsifiers; it does not report that any implementation, schema, product, test, or observation conforms or has been validated, frozen, or made ready.

Conformance reading guide

A future conformance review should trace every `SCI-AST-REQ-NNN` to an explicit representation, application point, failure path, persisted identity, and one or more independent falsification fixtures. It must evaluate the ordered composition as a whole, not merely compare output shape. It must also show that unavailable authority remains unavailable, that no stale or default value enters a required result, and that AST does not absorb ALIGN, RTC, MAP, geometry-producer, target-selection, or downstream-use ownership.

Review object	Minimum prospective evidence
Identity and parentage	Exact observation, occurrence, grid/slot, geometry, frame, center/WCS, plan/version, and lifecycle binding.
Numerical relation	Unit/topology checks, independent oracle, limiting cases, directional signs, domain and non-finite behavior.
Failure behavior	Typed unavailability, no substitute/default/stale state, and required-output propagation.
Ownership	No reconstruction or policy selection beyond AST's bounded TAN/WCS coordinate authority.
Parity	Deterministic sequential/parallel and supported real/simulation scientific identity.

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1 Notation and typed identities

This module fixes the symbols used by both views of the SCI-AST v0.1 authority. A symbol is meaningful only together with its declared identity, unit, frame, topology, support, parent, validity, and availability. Array position or a similar name is never an identity.

1.1 Indices and scientific objects

Symbol	Meaning
o	Immutable observation identity.
d	Admitted detector/acquisition occurrence, including its observation binding; it is not merely an array row or an optional design identity.
j	Stable slot on the exact admitted SCI-ALIGN detector-reference grid.
n	Stable output-sample identity on an exact resolved SCI-RTC output grid.
$k \in \{0, 1\}$	Producer-selected pointing-support record identity.
p	Pixel identity used by a MAP-owned deposition/gridding operator.
F	Declared output frame. H denotes the declared native horizon frame; J denotes the explicitly identified equatorial J2000 frame used by Science. The letters do not supply missing frame semantics.
$\boldsymbol{r}$	Unit direction in a declared spherical frame.
$\boldsymbol{e}_\ell, \boldsymbol{e}_b$	Local orthonormal tangent basis toward increasing longitude at fixed latitude and increasing latitude.
$\boldsymbol{c}_k, \boldsymbol{c}_j$	Pointing-support correction and its realization at aligned occurrence j, expressed as declared tangent components in radians.
$\boldsymbol{g}_d$	Selected detector-geometry datum, including its immutable APT realization and occurrence binding.
$\mathcal{G}_\gamma$	Selected, versioned geometry and field-rotation relation with lineage γ; it includes the declared gauge, basis, handedness, pivot state, affine limitations, time/elevation support, and uncertainty availability.
$\boldsymbol{\xi}_{dj}$	Realized detector tangent displacement supplied by $\mathcal{G}_\gamma$ at the current aligned sample time and elevation.
$\mathrm{R}_{F \leftarrow H, j}$	Named frame operator from the declared native horizon frame to F at occurrence j, including required time, site, epoch/equinox, EOP/refraction policy, version, and validity.
$\boldsymbol{w}$	Continuous tangent-plane coordinate, in arcseconds for Point/OOF/Beammap and degrees for Science.
$\boldsymbol{q} = (q_1, q_2)^{\mathrm{T}}$	Continuous one-based FITS pixel coordinate.
(i, j_{mem})	Zero-based in-memory row and column. The subscript on j_{mem} distinguishes it from an ALIGN-grid slot.
$\mathcal{W}_o$	Immutable realized WCS identity for the declared observation/product plan.
θ^{A}_{dj}	AST coordinate on the exact ALIGN-grid parent.
$\theta^{\mathrm{RTC}}_{dn}$	Separately parented AST coordinate on the exact resolved RTC output grid.
G_{pi}	Estimator-specific sample-to-pixel deposition/gridding operator owned by SCI-MAP, even when materialized by AST as a delegated service.
$J^{(\mathrm{out} \leftarrow \mathrm{in})}_{\mathrm{AST},0}$	Typed `astrometric_map_center_jacobian`, evaluated at the declared physical center.

1.2 Units, topology, and indexing

Angles used by the spherical equations are radians unless an equation states otherwise. Define

$$\kappa \equiv \frac{\pi}{180\,3600} \quad \text{radian per arcsecond.} \tag{1}$$

FITS/WCS pixel coordinates are one-based. In-memory rows and columns are zero-based. Internal windows and supports are half-open and retain stable identities. A longitude component must declare whether it is a coordinate increment or an on-sky tangent component.

For a period P, *shortest signed difference* means a deterministic member of $[-P/2, P/2)$, while persisted normalized longitude is a member of $[0, P)$. At a coordinate pole, the unit vector rather than longitude is the direction identity.

1.3 Exact parent identities

The parent of θ^{A}_{dj} is

$$\Pi^{A}_{dj} = (o, d, j,\ \text{ALIGN profile/grid/slot/source relation},\ \gamma,\ \text{observing state},\ \text{correction},\ F,\ \mathcal{W}_o). \tag{2}$$

The parent of θ^{RTC}_{dn} strictly extends, rather than relabels, that identity:

$$\begin{aligned}\Pi^{\text{RTC}}_{dn} = (o, d, n, &\ \Pi^{A}_{dM_n}, \text{RTC product/plan/grid}, M_n, t_n,\\ &\text{phase/delay}, \text{segment}, \text{decimation},\\ &\text{support/response/correction state}).\end{aligned} \tag{3}$$

Here M_n denotes the exact RTC-selected representative aligned occurrence when applicable; it is never selected by AST.

1.4 Availability and cause types

Every optional or unresolved quantity carries a typed state and exact reason. The vocabulary used here is:

- `available` and `available_conditional`;
- `unavailable_input`, `unavailable_authority`, and `unavailable_unsupported`; and
- `not_applicable` and `not_persisted_standard`.

A producer may refine these states, but must not collapse unavailable into numeric zero or an empty unlabeled field.

Five cause types are distinct throughout:

1. detector-occurrence identity;
2. detector signal-coordinate validity and numerical payload availability for x and r;
3. SCI-ALIGN mapping validity;
4. SCI-AST coordinate validity; and
5. downstream use-specific eligibility under a named owner's policy.

No cause is an alias, precedence rule, or rescue rule for another.

2 Normative scientific requirements

The key word "shall" marks a falsifiable v0.1 obligation. The identifiers are stable and sequential; a successor may append requirements but shall not renumber these identities.

2.1 Identity, input admission, and cause separation

`SCI-AST-REQ-001` — **One-way lifecycle.** Requested, effective, observation-resolved, and realized AST states shall be distinct one-way objects; a later state shall not rewrite the accepted request.

`SCI-AST-REQ-002` — **Observation identity.** Every realized AST bundle shall bind one exact immutable observation identity and product-plan identity.

`SCI-AST-REQ-003` — **Detector occurrence.** Every detector coordinate shall bind an exact detector/acquisition occurrence and its observation, not merely an array position or optional design identity.

`SCI-AST-REQ-004` — **Admitted detector binding.** A detector-to-geometry binding shall be either a proven observation-local row relation or an explicit keyed relation; bare row coincidence shall be rejected.

SCI-AST-REQ-005 — **Distinct detector identities.** Acquisition identity, selected measured-geometry identity, and optional design identity shall remain separate and shall not be substituted for one another.

SCI-AST-REQ-006 — **Immutable ALIGN transfer.** AST shall consume one exact compatible `SCI-ALIGN_TO_SCI-AST` transfer bundle without reconstructing, reinterpreting, or independently finalizing any transferred fact.

SCI-AST-REQ-007 — **No shaped substitute.** A missing required ALIGN payload, field, record, plan, version, source relation, or parent shall not be replaced by a similarly shaped or similarly named object.

SCI-AST-REQ-008 — **No clock or mapping reconstruction.** AST shall not reconstruct clocks, offsets, assignment, interpolation, gap/guard, continuity synthesis, epoch, topology, or source mapping from transferred values or array shape.

SCI-AST-REQ-009 — **Aligned-field typing.** Every consumed boresight or current observing-state field shall declare identity, registry version, unit, unchanged producer frame, topology, operator, exact source mapping/support, origin, validity, and uncertainty availability.

SCI-AST-REQ-010 — **Signal-independent coordinate availability.** A finite or requested x^A or r^A numerical detector-signal payload shall not be a prerequisite for an otherwise fully defined AST coordinate.

SCI-AST-REQ-011 — **Paired signal provenance.** Transferred x^A/r^A producer facts shall retain their common source relation and origin/synthesis state and their coordinate-specific numerical-validity and payload states without mixing or cross-filling.

SCI-AST-REQ-012 — **Five typed causes.** Detector-occurrence identity, signal-coordinate validity and payload availability, ALIGN mapping validity, AST coordinate validity, and downstream use-specific eligibility shall be represented as five distinct causes with no implicit precedence, alias, or rescue rule.

2.2 Pointing support and correction

SCI-AST-REQ-013 — **Producer selection.** A named pointing-support producer shall select the exact correction record identities and native support; AST shall not select a calibrator, replacement, or stale record.

SCI-AST-REQ-014 — **Correction semantics.** Each correction shall declare vector meaning, sign, ordered basis, coordinate-increment versus tangent meaning, arcsecond unit, parent, support mode/times, covariance availability, and application count before admission.

SCI-AST-REQ-015 — **Explicit sign adapter.** A producer residual with the opposite action shall be converted only by one explicit, versioned adapter whose direction and single application are recorded; an ambiguous sign shall fail before state replacement.

SCI-AST-REQ-016 — **Constant support.** One finite correction pair shall realize the `constant` mode of Equation 13 over its declared admitted observation support.

SCI-AST-REQ-017 — **Bracketed-MJD support.** Two finite correction pairs with finite, strictly increasing MJD supports shall realize Equation 14 only for occurrences satisfying $0 \le \lambda_j \le 1$.

SCI-AST-REQ-018 — **No extrapolation.** AST shall neither clamp, extrapolate, choose the nearest endpoint, nor reuse another observation for an occurrence outside producer-selected correction support.

SCI-AST-REQ-019 — **Legacy-span admission.** Two deliberately absent support times may select only the explicit `legacy_observation_span` mode, using the exact first and last admitted aligned sample identities/times in Equation 15.

SCI-AST-REQ-020 — **Ambiguous support failure.** Mixed absent/present, reversed, equal, non-finite, or otherwise ambiguous two-support encodings shall fail; a one-sample span is admissible only when both endpoint corrections are exactly equal and reduce to constant mode.

SCI-AST-REQ-021 — **Current-state separation.** Pointing interpolation shall use the exact ALIGN-admitted time relation, while field rotation shall use the aligned current science-sample time/elevation; the latter shall not be inferred from a pointing-support record.

SCI-AST-REQ-022 — **Support covariance.** When support covariance is reported, AST shall apply Equation 16, retain supplied cross-covariance, and keep model, selection, calibrator, and systematic terms separate; unavailable terms shall not become zero.

2.3 Detector geometry, field rotation, and composition

`SCI-AST-REQ-023` — **Geometry authorities.** Every selected geometry shall bind SCI-BEAM's measured relative-geometry authority and limitations plus an exact named observation-specific association or transformation authority and instance, or shall be typed unavailable.

`SCI-AST-REQ-024` — **Complete geometry artifact.** The selected artifact shall declare raw observation-local coordinates and, where applicable, horizon-fixed/derotated coordinates; ordered basis/handedness; units; gauge; angle support; versioned rotation law; pivot state; covariance; unresolved affine modes; exact APT realization; occurrence binding; and complete lineage.

`SCI-AST-REQ-025` — **No inferred origin or detector.** AST shall not infer APT origin equals telescope boresight, nominate a reference detector, or infer a geometry relation from design identity or row order.

`SCI-AST-REQ-026` — **Gauge preservation.** Every geometry realization shall preserve the declared origin/recentering translation, basis, handedness, tangent versus coordinate-increment meaning, and conventional pivot parent.

`SCI-AST-REQ-027` — **Current field rotation.** AST shall evaluate the selected versioned field-rotation relation at the current aligned occurrence time/elevation and other declared current state, within its exact support.

`SCI-AST-REQ-028` — **Rotation support.** A geometry or rotation value outside its declared time, elevation, or domain support shall be unavailable; AST shall not clamp, extrapolate, or replace it with an unversioned field-rotation relation.

`SCI-AST-REQ-029` — **Affine and pivot availability.** Unknown pivot covariance or unresolved translation, scale, skew, shear, or other affine terms shall retain typed unavailability/bounds and shall not be set silently to zero or identity.

`SCI-AST-REQ-030` — **Same-realization key.** Every pointing-support observation and the science observation receiving its correction shall have identical K_γ from Equation 6, unless Requirement 031 is satisfied.

`SCI-AST-REQ-031` — **Authorized equivalence transform.** A cross-realization transform shall name both immutable parents, prove equivalence over a declared detector/ time/elevation domain, act in the correct direction and units, propagate covariance, cross-covariance and transform uncertainty, and record authority, version, validity, and failure behavior.

`SCI-AST-REQ-032` — **Incompatible transfer failure.** Absent the complete transform in Requirement 031, any mismatch in K_γ shall make the dependent correction/coordinate unavailable without rewriting unrelated ALIGN or signal facts.

`SCI-AST-REQ-033` — **Noncommuting order.** AST shall apply the eight operations in Section 3.5 in their stated order and retain each stage's exact parents.

`SCI-AST-REQ-034` — **No duplicate operation.** Composition history and application counts shall prevent any pointing, geometry, rotation, frame, RTC, or upstream coordinate correction from being applied twice.

`SCI-AST-REQ-035` — **Bounded small-angle operator.** Production use of Equation 21 shall require a preregistered field/elevation/ time domain, detector population, coordinate and pixel metric, named use, tolerance owner/value, spherical oracle/version, and fail-closed behavior; AST shall invent neither a universal threshold nor an automatic fallback.

2.4 Direction, frame, topology, and projection

`SCI-AST-REQ-036` — **Vector direction identity.** A sky direction shall be represented by a finite normalizable unit vector in one declared frame; longitude alone shall not establish direction at a pole.

`SCI-AST-REQ-037` — **Longitude-component type.** Every longitude-like correction or offset shall declare coordinate-increment or on-sky tangent meaning and shall be normalized according to Equation 11.

`SCI-AST-REQ-038` — **Circular topology.** Longitude differences shall use Equation 8; persisted longitude shall use Equation 9 in its declared unit interval.

`SCI-AST-REQ-039` — **Product-family frame split.** Point, OOF, and Beammap shall use native AltAz tangent azimuth/elevation coordinates in arcseconds; Science shall use explicitly identified equatorial J2000 TAN coordinates in degrees.

`SCI-AST-REQ-040` — **Named frame transform.** Every non-identity frame operation shall name source/target frames, epoch/equinox, time scale, site, model/version, EOP/refraction applicability, required inputs, validity, and uncertainty availability.

`SCI-AST-REQ-041` — **No frame relabeling.** Missing or incompatible frame authority shall make the dependent coordinate unavailable; AST shall not equate J2000, FK5, ICRS, apparent, observed, or unrefracted frames by name or default.

`SCI-AST-REQ-042` — **Exact TAN domain.** The forward TAN mathematical domain shall be finite $D > 0$ as in Equation 22; v0.1 shall impose no universal positive $D_{\min}$.

`SCI-AST-REQ-043` — **TAN invalidity.** A non-finite direction, D, tangent output, or $D \leq 0$ shall produce explicit AST coordinate invalidity and no stale, center, clamped, or edge-pixel substitute.

`SCI-AST-REQ-044` — **Projection units and axes.** Tangent outputs shall preserve the exact frame, ordered axes, topology, and units in Equations 25 and 26.

2.5 Physical center, WCS, and pixel facts

`SCI-AST-REQ-045` — **Center selection authority.** Every WCS shall bind the exact physical center and selection/product-plan identity supplied by a named upstream target/product authority.

`SCI-AST-REQ-046` — **No center inference.** AST automatic WCS shall never search, fit, recenter, derive a center from map bounds, or default a missing/ambiguous center; the affected WCS shall be unavailable.

`SCI-AST-REQ-047` — **Legacy all-zero adapter.** Only the exact all-zero pattern across all six named legacy CRPIX/CRVAL/TAN controls shall adapt to internal `automatic`; the adapter and original request shall be recorded.

`SCI-AST-REQ-048` — **Unsupported explicit WCS.** A nonzero or non-finite legacy explicit WCS control request shall fail at v0.1 admission as unsupported; numeric zero outside the exact six-field adapter shall remain an ordinary value.

`SCI-AST-REQ-049` — **Immutable WCS identity.** The realized WCS shall bind projection, center authority, CRVAL, one-based CRPIX, CD or equivalent PC+CDELT, axis order/sign/rotation/handedness, dimensions/bounds, units, frame/epoch, precision, product plan, and version.

`SCI-AST-REQ-050` — **Nonsingular continuous WCS.** The CD-equivalent matrix shall be finite and nonsingular, and continuous FITS coordinates shall satisfy Equation 27; hidden transpose, flip, or axis exchange shall fail directional checks.

`SCI-AST-REQ-051` — **Reference-pixel policy.** When the center policy is geometric array center, CRPIX shall satisfy Equation 28; any other declared policy shall still place CRVAL exactly at its one-based CRPIX and shall be recorded.

`SCI-AST-REQ-052` — **Index bases.** FITS pixel centers and zero-based memory indices shall obey Equation 29; a consumer shall not infer one convention from the other.

`SCI-AST-REQ-053` — **Nominal pixel rule.** When requested, the v0.1 nominal containing pixel shall obey the half-open nearest-center rule in Equation 30, or shall name a separately versioned rule with equivalent boundary tests.

`SCI-AST-REQ-054` — **Bounds separation.** Continuous coordinate validity, nominal discrete pixel identity, in-bounds/out-of-bounds state, MAP support/coverage, and coordinate precision shall remain distinct facts.

`SCI-AST-REQ-055` — **Non-finite pixel behavior.** A non-finite continuous coordinate or pixel shall be invalid and shall never be cast to an integer, mapped to an edge, or retained from a prior occurrence.

2.6 Validity, atomic output, products, and lifecycle

`SCI-AST-REQ-056` — **Dedicated AST validity.** Every coordinate role shall publish a dedicated AST coordinate-validity cause and exact reason, distinct from ALIGN and downstream states.

`SCI-AST-REQ-057` — **Dependency-limited failure.** A missing or invalid required input shall make only its dependent coordinate/product/claim unavailable unless the required product contract demands whole-product failure; it shall not rewrite unrelated exact relations.

`SCI-AST-REQ-058` — **Atomic bundle.** Direction, tangent and continuous pixel coordinates, WCS, detector/grid parents, validity, uncertainty/Jacobian availability, and provenance shall be published atomically for each required role.

`SCI-AST-REQ-059` — **Required-output failure.** Failure to construct or write a required coordinate, WCS, provenance, or product shall fail the required product and reduction; partial or mislabeled output shall not be success.

`SCI-AST-REQ-060` — **Typed optional absence.** An optional, unsupported, or unavailable coordinate, Jacobian, or G_{pi} product shall carry its exact typed state and reason and shall not masquerade as a complete bundle.

`SCI-AST-REQ-061` — **Full and mini TOD truth.** Full TOD may retain coordinate arrays with compact interpretation/validity state; mini TOD shall declare coordinate arrays unavailable rather than silently omit their meaning.

`SCI-AST-REQ-062` — **Complete replacement.** Observation resolution shall build a complete immutable candidate before atomic replacement; failure shall leave no partial or prior AST state usable by the next observation.

`SCI-AST-REQ-063` — **Sequential/parallel equivalence.** Under the declared numerical policy, sequential and parallel schedules shall produce identical per-parent coordinates, nominal pixels, validity, counts, WCS identity, and deterministic provenance order.

`SCI-AST-REQ-064` — **Real/simulation parity.** Supported real and simulation paths shall use the same equations, support, wrap, index, validity, and output-failure rules; simulation shall differ only through explicit origin/generator provenance.

2.7 Map-center Jacobian and uncertainty

`SCI-AST-REQ-065` — **Typed Jacobian schema.** Every available `astrometric_map_center_jacobian` shall bind exact input domain, output codomain, direction, axis order, units, center/WCS/product parents, family, approximation domain, method/version, and included/unavailable terms.

`SCI-AST-REQ-066` — **Family request gate.** The map-center Jacobian shall be typed unavailable until a named product family requests its exact domain, codomain, axis order, units, and direction.

`SCI-AST-REQ-067` — **Center-only scope.** V0.1 shall evaluate the Jacobian only at the declared map center and shall not create per-sample, per-detector, per-pixel, response-grid, dense, or broadly composed astrometric response/covariance.

`SCI-AST-REQ-068` — **Response exclusions.** The AST Jacobian shall not be labeled or used as a source-amplitude, beam, RTC temporal, MAP transfer, or point-source response.

`SCI-AST-REQ-069` — **Available covariance only.** AST shall transform only declared available map-center covariance and cross-covariance terms using the typed realized derivative; unavailable terms shall remain unavailable.

`SCI-AST-REQ-070` — **No unjustified total.** A total map-center covariance shall be formed only when all claimed components, cross-covariances, conditioning, and combination assumptions are declared.

`SCI-AST-REQ-071` — **Native tangent uncertainty.** When native tangent uncertainty is the product quantity, AST shall preserve it with its frame, basis, units, parent, conditioning, and availability rather than relabel it as another domain.

`SCI-AST-REQ-072` — **Selection uncertainty.** Discrete selection/availability uncertainty shall not be converted to Gaussian covariance without a separately approved model and parent.

2.8 ALIGN-grid, RTC-grid, and MAP handoff

`SCI-AST-REQ-073` — **ALIGN-grid coordinate parent.** θ^{A}_{dj} shall bind the exact ALIGN profile, plan, grid, slot, detector occurrence, source relation, geometry, observing state, correction, frame, and WCS parents in Equation 2.

`SCI-AST-REQ-074` — **RTC-grid coordinate parent.** $\theta^{\mathrm{RTC}}_{dn}$ shall bind the exact RTC product/plan/grid and all facts in Equation 3, even when it numerically equals a phase-zero representative ALIGN occurrence.

`SCI-AST-REQ-075` — **RTC fact completeness.** The RTC parent shall include owner, product, plan, version, detector and output sample, representative occurrence, selected output time, scale, support, uncertainty, phase/delay sign, unit, value and count, segment, decimation factor, phase and support, correction state, temporal response, and status/reasons.

SCI-AST-REQ-076 — No RTC reconstruction. AST shall not infer or reconstruct RTC plan, grid, cadence, phase, delay, time shift, representative occurrence, segment, decimation support, response, or correction state from shape or numerical coincidence.

SCI-AST-REQ-077 — No repeated upstream operation. Constructing $\theta_{dn}^{\mathrm{RTC}}$ shall not repeat ALIGN, reapply an RTC/upstream coordinate correction, or silently reuse pre-RTC time.

SCI-AST-REQ-078 — Full RTC sky-response support. A response-aware consumer shall retain the RTC temporal operator and the full contributing set of ALIGN-grid coordinates; it shall not replace them by one nominal output coordinate.

SCI-AST-REQ-079 — No angular signal filtering. AST shall not apply RTC signal coefficients directly to angular coordinates as a representation of filtered sky response, consistent with Equation 37.

SCI-AST-REQ-080 — Projection ownership. AST shall own TAN/WCS astrometric projection and continuous coordinate facts; MAP shall own sample-to-pixel deposition/ gridding selection, numerical operator, support, normalization, boundaries, conservation, response, covariance, accumulation, cropping, and reprojection.

SCI-AST-REQ-081 — Delegated materialization gate. AST may materialize geometry-only or complete G_{pi} only for an exact MAP-owned request naming kernel family, parameters, support, normalization, edge/boundary behavior, conservation claim, requested artifact form, response/covariance role, plan identity, and version.

SCI-AST-REQ-082 — Delegated parentage. Every materialized G_{pi} shall bind the exact AST coordinate parent, WCS/pixel geometry, exact RTC output-grid parent when applicable, and exact MAP projection-plan parent as in Equation 38.

SCI-AST-REQ-083 — No ownership transfer. AST materialization shall neither select nor authorize estimator-specific G_{pi} and shall not transfer MAP scientific ownership to AST.

2.9 Provenance, availability, and verification obligations

SCI-AST-REQ-084 — Requested provenance. Requested provenance shall retain exact accepted values and source identities, pointing pairs/support, center, projection, scale, orientation, dimensions, products, and inactive expert values.

SCI-AST-REQ-085 — Effective provenance. Effective provenance shall retain activation, normalized units/topology, sign adapter, support policy, WCS request state, projection family, compatibility decisions, and their versions.

SCI-AST-REQ-086 — Observation-resolved provenance. Observation-resolved provenance shall retain observation/detector/geometry/ALIGN/RTC identities as applicable, support bracket, frame operator, field origins/validity, center authority, exact WCS/bounds, approximation record, and uncertainty availability.

SCI-AST-REQ-087 — Realized provenance. Realized provenance shall retain applied support and endpoint identities, occurrence and product counts, exclusions and reasons, coordinate extrema/wrap events, exact serialized WCS, algorithm and contract versions, unavailable terms, required-output outcomes, product links, and artifact digests.

SCI-AST-REQ-088 — Reconstructible compactness. Compact provenance may reuse stable ALIGN-grid and detector bindings, but shall reconstruct every policy decision and bind each persisted coordinate/product to its exact parents and versions.

SCI-AST-REQ-089 — Empirical gates. Small-angle, time-representation, and large-array precision adequacy shall be tested against preregistered domains, metrics, tolerances, owners, oracles, and failure rules; a required-data skip shall not be counted as a pass.

SCI-AST-REQ-090 — Claim separation. Scientific-contract authorship, implementation conformity, representation fidelity, empirical validation, observational performance, freeze, and production readiness shall remain separate claims; this Stage B draft shall assert none of the latter claims.

3 Scientific definitions and ownership boundary

3.1 Bounded scientific object

For an admitted observation, detector occurrence, and explicitly parented sample, SCI-AST realizes an atomic coordinate record

$$\mathcal{A} = \{\widehat{\boldsymbol{r}}^F, \boldsymbol{w}, \boldsymbol{q}, \nu_{\mathrm{AST}}, \mathcal{W}_o, \mathcal{U}, \Pi, \mathcal{P}\}, \tag{4}$$

where the direction, tangent coordinate, continuous pixel coordinate, coordinate-validity cause, WCS, bounded uncertainty/Jacobian availability, complete parentage, and four-stage provenance are mutually consistent. Optional nominal containing-pixel and zero-based memory-index facts may be attached under an exact rounding convention. A finite number without this identity is not an AST coordinate.

The v0.1 estimator begins with the exact immutable SCI-ALIGN transfer bundle, producer-selected pointing support, selected observation-specific detector geometry, declared frame plan, and upstream-selected physical center/product plan. It ends with the coordinate/WCS bundle. It does not select a calibrator or center, create detector geometry, reconstruct ALIGN or RTC, filter detector signals, accumulate a map, fit a source, infer a beam or pointing solution, calibrate signal, or decide a universal use policy.

3.2 Exact ALIGN import

Here *detector-reference* means the selected detector-stream reference interface, its clock relation, and its assigned grid. It never means a reference detector. A nominal slot or nominal time alone is not proof of a physical event, physical integration, or acquired exposure.

The SCI-ALIGN input is one immutable transfer, not a menu. It binds the detector-reference interface, grid and slot identities; native, corrected and assigned times; native source occurrences and temporal weights; residuals; field identity, units, unchanged producer frame, topology and operator; origin/synthesis/unavailability and acquired-exposure facts; interval and support identities; validity; response and uncertainty availability; and the requested, effective, observation-resolved, and realized ALIGN plan and provenance. Expanded source detail may be replaced only by an exactly reconstructible generative record supplied by ALIGN.

The paired detector-signal source relation is

$$x^A = A_{\mathrm{ALIGN}} x^{\mathrm{acq}}, \qquad r^A = A_{\mathrm{ALIGN}} r^{\mathrm{acq}}, \tag{5}$$

with identical native occurrences, weights, residuals, slots, common source relation, and origin/synthesis state. Numerical validity and payload availability remain coordinate-specific. AST needs the occurrence/time/ mapping and observing-state relation, not a finite or requested x^A or r^A value. No relation crosses a Tune/readout mapping-revision boundary without a separately named authority, response, and uncertainty.

Aligned boresight, current science-sample time, current elevation, current azimuth when required, and all other declared occurrence-local observing-state fields are separate from pointing-correction records. Field rotation consumes the current aligned observing state and never infers current elevation from a bracketing correction record.

3.3 Pointing-support modes

A named producer selects the exact correction records and their native support. ALIGN supplies their admitted relation to detector-reference time. AST admits exactly one of three modes:

`constant`
: one finite correction pair over the declared observation support;

`bracketed_mjd`
: two finite correction pairs with finite, strictly increasing MJD support that brackets every dependent occurrence;

`legacy_observation_span`
: two correction pairs whose two support times are both deliberately absent at the documented legacy boundary, interpolated only over the exact first and last admitted aligned sample identities/times.

Mixed, ambiguous, equal, reversed, non-finite, or unsupported time encodings do not choose a mode. AST interpolates within the selected mode and never extrapolates, clamps, chooses a nearest record, or reuses a stale observation.

3.4 Detector geometry and field rotation

SCI-BEAM owns measured relative detector geometry and its measurement limitations. An exact named TolProj/TolAPT or other approved authority owns the observation-specific association/transformation and immutable APT realization. AST consumes and composes the selected artifact; it neither creates focal-plane truth nor assigns generic geometry-origin authority to SCI-CAL.

The selected artifact declares, as applicable, raw observation-local and horizon-fixed/derotated coordinates; ordered basis and handedness; tangent or coordinate-increment meaning; angular units; origin/recentering gauge; effective angle and time/elevation support; versioned field-rotation law; conventional pivot with covariance or typed unavailability; geometry covariance and cross terms; unresolved translation, scale, skew, shear, or other affine modes; exact APT content identity; occurrence binding; and full association/transformation lineage. AST does not infer that APT origin is telescope boresight, nominate a reference detector, identify rows by bare coincidence, or reduce the physical relation to a pure rotation about a known pivot.

Pointing-support observations and the science observation receiving their correction have the identical compatibility key

$$\begin{aligned} K_\gamma = (\ &\text{APT realization}, \text{coordinate realization/gauge}, \text{geometry version}, \\ &\text{rotation law/version}, \text{pivot definition}, \text{AST composition convention}). \end{aligned} \tag{6}$$

A mismatch is incompatible unless a separately authorized exact transform names both immutable parents, proves equivalence over a declared domain, acts in the correct direction and units, propagates covariance, cross- covariance and transform uncertainty, and records authority, version, validity, and failure behavior.

3.5 Ordered, noncommuting AST composition

The canonical order is:

1. admit the exact aligned boresight and current observing state;
2. interpret and apply the producer-selected pointing correction inside its selected native support;
3. realize the selected detector geometry and field rotation at the current aligned sample time and elevation;
4. compose the detector direction under the declared sign, basis, handedness, origin, gauge, and pivot conventions;
5. apply the declared native or named frame transform;
6. apply circular topology and the tangent-plane projection;
7. realize the exact WCS from the upstream-selected center/product plan;
8. publish continuous FITS pixels and any explicitly requested nominal discrete pixel facts with their bounds state.

These operations are generally noncommuting. They are not reordered because a zero or infinitesimal special case appears to commute. Composition history prevents any already-applied operation from being applied twice. Delegated G_{pi} materialization is downstream of this order and is not a ninth AST composition stage.

3.6 Frame and product-family identities

Point, OOF, and Beammap use native AltAz tangent-plane azimuth/elevation coordinates in arcseconds. Science uses explicitly identified equatorial J2000 TAN coordinates in degrees. J2000, FK5, ICRS, apparent place, observed AltAz, and unrefracted AltAz are not synonyms. A transform declares source and target frames, epoch/equinox, time scale, site, version, required EOP or refraction inputs and their applicability. A missing transform is not repaired by relabeling or by inventing a default epoch.

Only nonpolarimetric Stokes-I use is in scope. A supplied HWPR timing fact may remain in the ALIGN parent, but AST v0.1 does not interpret polarization. Supported AltAz and RA/Dec simulations use the same applicable contract as real data; Galactic or other unsupported frame requests fail at admission.

3.7 Center, WCS, pixels, and request state

The physical center is selected by a named upstream target/product authority. AST realizes and encodes that exact selection. It never searches, fits, silently recenters, derives a center from bounds, or supplies a default.

The scientific WCS request states are `automatic`, `explicit`, `unsupported`, and `unavailable`. The exact all-zero pattern across the six legacy controls `crpix1`, `crpix2`, `crval1_J2000`, `crval2_J2000`, `tan_ra`, and `tan_dec` adapts only at the legacy boundary to `automatic`. Nonzero or non-finite legacy explicit requests are unsupported in v0.1 and fail at admission. Numeric zero everywhere else is an ordinary value.

An immutable WCS binds projection, physical-center authority, CRVAL, one-based CRPIX, CD or an exactly equivalent PC+CDELT representation, axis order/sign/rotation/handedness, dimensions, bounds, output units, frame/epoch, precision, product plan, and realization version. Continuous WCS coordinate, nominal containing pixel, zero-based memory index, in-bounds state, map coverage, and coordinate precision are distinct facts.

3.8 AST projection versus MAP deposition

SCI-AST owns the TAN/WCS astrometric projection: the exact geometric relation from an admitted detector-time occurrence to detector sky direction, tangent-plane coordinate, continuous FITS pixel coordinate, WCS identity, coordinate validity and bounded astrometric uncertainty, plus an optional nominal containing pixel under a declared rule.

SCI-MAP owns the estimator-specific sample-to-pixel deposition/gridding projection and realized G_{pi}: kernel family and parameters, support, one-hot or fractional behavior, normalization, edge/boundary treatment, conservation claim, numerical contribution, and response/covariance consequences. MAP also owns accumulation, support, weighting, coaddition, cropping, reprojection, and estimator/final-map validity.

AST may materialize a geometry-only or complete G_{pi} solely for an exact MAP-owned request. The artifact binds its AST coordinate and WCS parent, its exact RTC output-grid parent when applicable, and its MAP projection-plan identity. Delegated materialization transfers neither policy nor scientific ownership to AST.

3.9 ALIGN-grid and RTC-output-grid coordinates

θ^{A}_{dj} and $\theta^{\mathrm{RTC}}_{dn}$ are different scientific roles. The first is parented to the exact ALIGN grid and occurrence. The second is parented additionally to the exact RTC product, resolved plan, output grid, representative occurrence, selected output time, phase/delay convention, segment, decimation factor and phase, support, conditional response, and coordinate-correction state. Even a phase-zero RTC case that numerically selects one aligned occurrence retains the RTC parent and is not relabeled.

For an RTC signal operator, the full sky-response support jointly retains the temporal operator and all contributing θ^{A}_{dj}. Applying temporal signal coefficients directly to circular or nonlinear angular coordinates is not the filtered sky response, and one nominal output coordinate does not replace that support.

3.10 Map-center Jacobian and uncertainty

The only v0.1 astrometric response product is the typed `astrometric_map_center_jacobian` at the declared center. An available instance binds exact domain, codomain, direction, axis order, units, center/WCS/product parents, applicable family, approximation domain, method and version, and every included or unavailable uncertainty term. It is unavailable until a product family requests the exact domain and codomain.

The object is an astrometric coordinate derivative, not source-amplitude, beam, RTC temporal, MAP transfer, or point-source response. V0.1 does not publish per-sample, per-detector, per-pixel, response-grid, dense, or broadly composed covariance. Missing inputs and cross terms remain unavailable; a total is formed only under complete declared components and combination assumptions.

3.11 Lifecycle, products, and claim separation

Requested, effective, observation-resolved, and realized state are distinct one-way objects. Observation resolution constructs a complete immutable candidate before atomic replacement of prior state. Failure or absence resets dependent optional state and leaves no stale candidate usable by another observation. Sequential and parallel scheduling, and supported real and simulation paths, preserve the same coordinate meaning and deterministic identity.

Full TOD may retain coordinate arrays with compact interpretation and validity state. Mini TOD declares coordinate arrays unavailable. A required coordinate, WCS, provenance, or output failure fails the required product and reduction; partial or mislabeled output is not success.

This document defines a scientific contract and falsifiable predictions. It does not itself establish implementation conformity, representation fidelity, empirical adequacy, observational performance, validation, freeze, production readiness, or scientific approval of this Stage B draft.

4 Canonical equations and typed operators

The exact spherical equations in this section are mathematical identities and the independent offline oracle. The v0.1 production geometry may retain the established small-angle operator only inside the preregistered adequacy domain defined in Section 5; no automatic spherical fallback or universal numerical threshold is implied.

4.1 Directions, tangent basis, and circular topology

For longitude ℓ and latitude b in a declared frame,

$$\boldsymbol{r}(\ell, b) = \begin{bmatrix} \cos b \cos \ell \\ \cos b \sin \ell \\ \sin b \end{bmatrix}, \quad \boldsymbol{e}_\ell = \begin{bmatrix} -\sin \ell \\ \cos \ell \\ 0 \end{bmatrix}, \quad \boldsymbol{e}_b = \begin{bmatrix} -\sin b \cos \ell \\ -\sin b \sin \ell \\ \cos b \end{bmatrix}. \tag{7}$$

Away from a coordinate pole, $\boldsymbol{e}_\ell$ and $\boldsymbol{e}_b$ are unit tangent vectors toward increasing longitude at fixed latitude and increasing latitude. The direction remains $\boldsymbol{r}$ at a pole.

For period P,

$$\mathrm{wrap}_P(\theta) = \theta - P \left\lfloor \frac{\theta + P/2}{P} \right\rfloor \in [-P/2, P/2), \tag{8}$$

$$\mathrm{norm}_P(\theta) = \theta - P \left\lfloor \frac{\theta}{P} \right\rfloor \in [0, P). \tag{9}$$

Longitude and azimuth use $P = 2\pi$. The half-open endpoint convention is deterministic.

4.2 Exact displacement oracle and input normalization

For tangent components $\boldsymbol{\eta} = (\eta_\ell, \eta_b)^{\mathrm{T}}$, let $\rho = \|\boldsymbol{\eta}\|$ and, for $\rho > 0$, $\boldsymbol{u} = (\eta_\ell \boldsymbol{e}_\ell + \eta_b \boldsymbol{e}_b)/\rho$. The exact spherical displacement oracle is

$$\mathrm{Exp}_{\boldsymbol{r}}(\boldsymbol{\eta}) = \begin{cases} \cos \rho \, \boldsymbol{r} + \sin \rho \, \boldsymbol{u}, & \rho > 0, \\ \boldsymbol{r}, & \rho = 0. \end{cases} \tag{10}$$

Renormalization may remove roundoff only; it may not hide invalid input.

An input longitude-like component is normalized at the boundary as

$$\eta_\ell = \begin{cases} \cos b \, \Delta\ell, & \text{declared coordinate increment}, \\ \Delta\ell_{\mathrm{tangent}}, & \text{declared on-sky tangent component}. \end{cases} \tag{11}$$

The declared correction pair $\boldsymbol{a}_k$ in arcseconds becomes

$$\boldsymbol{c}_k = \kappa \boldsymbol{a}_k. \tag{12}$$

Positive components move toward the increasing declared tangent-basis axes. An upstream residual requiring the opposite action needs one explicit, parented sign adapter and an application count of one.

4.3 Pointing-support realization

One finite pair is constant:

$$\boldsymbol{c}_j = \boldsymbol{c}_0. \tag{13}$$

For two finite, strictly increasing MJD supports $m_0 < m_1$ and aligned occurrence time m_j,

$$\lambda_j = \frac{m_j - m_0}{m_1 - m_0}, \qquad \boldsymbol{c}_j = (1 - \lambda_j)\boldsymbol{c}_0 + \lambda_j \boldsymbol{c}_1, \qquad 0 \le \lambda_j \le 1. \tag{14}$$

For explicit legacy observation-span mode, where t_- and t_+ are the times of the exact first and last admitted aligned sample identities,

$$\lambda_j^{\mathrm{span}} = \frac{t_j - t_-}{t_+ - t_-}, \qquad \boldsymbol{c}_j = (1-\lambda_j^{\mathrm{span}})\boldsymbol{c}_0 + \lambda_j^{\mathrm{span}}\boldsymbol{c}_1, \quad t_- \le t_j \le t_+. \tag{15}$$

A one-sample span is defined only when the endpoint corrections are exactly equal, in which case Equation 13 applies. When the two-support covariance is available, stack $\boldsymbol{c}_* = (\boldsymbol{c}_0^{\mathrm{T}}, \boldsymbol{c}_1^{\mathrm{T}})^{\mathrm{T}}$ and define

$$\mathrm{H}_j = \begin{bmatrix}(1-\lambda_j)\mathrm{I}_2 & \lambda_j \mathrm{I}_2\end{bmatrix}, \quad \mathrm{C}_{c,j}^{\mathrm{formal}} = \mathrm{H}_j \mathrm{C}_{c,*} \mathrm{H}_j^{\mathrm{T}}. \tag{16}$$

Cross-covariance is retained when supplied. Model, selection, and systematic terms remain separately labeled.

4.4 Ordered detector-direction composition

Let $\boldsymbol{r}_j^B$ be the exact aligned boresight in the declared native horizon frame. The spherical oracle for the first two geometric stages is

$$\begin{aligned}
\boldsymbol{r}_j^{H,c} &= \mathrm{Exp}_{\boldsymbol{r}_j^B}(\boldsymbol{c}_j), &(17)\\
\boldsymbol{\xi}_{dj} &= \mathcal{G}_\gamma(\boldsymbol{g}_d; t_j, E_j, \text{declared occurrence-local state}), &(18)\\
\boldsymbol{r}_{dj}^H &= \mathcal{C}_\gamma(\boldsymbol{r}_j^{H,c}, \boldsymbol{\xi}_{dj}), &(19)\\
\boldsymbol{r}_{dj}^F &= \mathrm{R}_{F\leftarrow H,j}(\boldsymbol{r}_{dj}^H). &(20)
\end{aligned}$$

$\mathcal{C}_\gamma$ is the selected composition relation and carries the same gauge, pivot, basis, handedness, affine limitations, and lineage as $\mathcal{G}_\gamma$. In the admitted pure tangent-displacement limiting case, $\mathcal{C}_\gamma(\boldsymbol{r}, \boldsymbol{\xi}) = \mathrm{Exp}_{\boldsymbol{r}}(\boldsymbol{\xi})$; the contract does not presume that this limiting case describes every selected artifact.

The established production small-angle relation may be written

$$\boldsymbol{\eta}_{dj}^{\mathrm{prod}} \simeq \boldsymbol{c}_j + \mathrm{B}_{dj}\boldsymbol{f}_d, \tag{21}$$

only when B_{dj} is the declared selected geometry/rotation mapping and the preregistered adequacy record admits the exact occurrence, detector, product use, and error metric. Equation 10 and the full selected geometry relation are the offline oracle for that record.

4.5 Gnomonic projection

For output direction $\boldsymbol{r}$, center $\boldsymbol{r}_0$, and center basis $(\boldsymbol{e}_1, \boldsymbol{e}_2)$,

$$D = \boldsymbol{r}\cdot\boldsymbol{r}_0, \qquad x = \frac{\boldsymbol{r}\cdot\boldsymbol{e}_1}{D}, \qquad y = \frac{\boldsymbol{r}\cdot\boldsymbol{e}_2}{D}, \qquad D > 0. \tag{22}$$

The inverse is

$$\boldsymbol{r} = \frac{\boldsymbol{r}_0 + x\boldsymbol{e}_1 + y\boldsymbol{e}_2}{\sqrt{1+x^2+y^2}}. \tag{23}$$

In longitude and latitude, with $\Delta\ell = \mathrm{wrap}_{2\pi}(\ell - \ell_0)$,

$$\begin{aligned}
&D = \sin b_0 \sin b + \cos b_0 \cos b \cos\Delta\ell,\\
&x = \frac{\cos b \sin\Delta\ell}{D}, \qquad y = \frac{\cos b_0 \sin b - \sin b_0 \cos b \cos\Delta\ell}{D}.
\end{aligned} \tag{24}$$

The mathematical domain is finite $D > 0$. There is no universal positive $D_{\min}$. A separately preregistered footprint bound may be tighter, but does not alter the mathematical domain. Non-finite input or output and $D \le 0$ are coordinate invalidity, never a center or edge pixel.

For Point/OOF/Beammap,

$$\boldsymbol{w} = 206264.806247\,(x, y)^{\mathrm{T}} \text{ arcsec}, \tag{25}$$

with ordered AZOFFSET/ELOFFSET identity. For Science,

$$\boldsymbol{w} = \frac{180}{\pi}(x, y)^{\mathrm{T}} \text{ degree}, \tag{26}$$

with explicitly identified RA—TAN/DEC–TAN axes in the declared equatorial J2000 frame.

4.6 Continuous WCS and nominal discrete pixel

Let $\boldsymbol{q}_0 = \mathrm{CRPIX}$ be one-based and let M be the declared nonsingular CD matrix in plane-units per pixel:

$$\boldsymbol{w} = \mathrm{M}(\boldsymbol{q} - \boldsymbol{q}_0), \qquad \boldsymbol{q} = \boldsymbol{q}_0 + \mathrm{M}^{-1}\boldsymbol{w}. \tag{27}$$

When the product plan places the projection center at the geometric array center for N_1 columns and N_2 rows,

$$\mathrm{CRPIX}_1 = \frac{N_1 + 1}{2}, \qquad \mathrm{CRPIX}_2 = \frac{N_2 + 1}{2}. \tag{28}$$

Even dimensions therefore have half-integer reference pixels. A different declared reference-pixel policy remains valid only when Equation 27 places CRVAL there exactly and provenance records that policy.

At pixel centers,

$$q_1 = j_{\mathrm{mem}} + 1, \qquad q_2 = i + 1. \tag{29}$$

The v0.1 half-open nearest-center assignment is

$$j_{\mathrm{mem}} = \left\lfloor q_1 - \tfrac{1}{2} \right\rfloor, \qquad i = \left\lfloor q_2 - \tfrac{1}{2} \right\rfloor, \quad 0 \le j_{\mathrm{mem}} < N_1, \quad 0 \le i < N_2. \tag{30}$$

Another rule is a separately identified operator with equivalent boundary tests; it is never inferred from integer casting.

4.7 Typed map-center Jacobian and covariance

For Equation 22, let $a = \boldsymbol{r} \cdot \boldsymbol{e}_1$ and $b = \boldsymbol{r} \cdot \boldsymbol{e}_2$. The direction-to-plane derivative is

$$\mathrm{J}_{w \leftarrow r} = \begin{bmatrix} (D\boldsymbol{e}_1^{\mathrm{T}} - a\boldsymbol{r}_0^{\mathrm{T}})/D^2 \\ (D\boldsymbol{e}_2^{\mathrm{T}} - b\boldsymbol{r}_0^{\mathrm{T}})/D^2 \end{bmatrix}, \tag{31}$$

followed by the declared unit conversion. For a family-requested typed input $\boldsymbol{c}_{\mathrm{in}}$ and output $\boldsymbol{c}_{\mathrm{out}}$,

$$J_{\mathrm{AST},0}^{(\mathrm{out} \leftarrow \mathrm{in})} = \left. \frac{\partial \boldsymbol{c}_{\mathrm{out}}}{\partial \boldsymbol{c}_{\mathrm{in}}} \right|_{\boldsymbol{r}_0, \mathcal{W}_o}. \tag{32}$$

Its direction, axes, units, family, center/WCS parent, approximation domain, method, and included or unavailable terms are part of the type. For an available map-center covariance term k,

$$\mathrm{C}_{\mathrm{out}}^{(k)} = J_{\mathrm{out} \leftarrow k} \mathrm{C}_k J_{\mathrm{out} \leftarrow k}^{\mathrm{T}}. \tag{33}$$

Supplied cross terms are propagated. No total is reported unless all claimed components, cross-covariances, and combination assumptions are declared.

4.8 Grid roles, RTC response, and delegated MAP operator

On the ALIGN grid and the RTC output grid, respectively,

$$\theta_{dj}^{A} = \mathcal{A}_{\mathrm{AST}}(d, j; \Pi_{dj}^{A}), \tag{34}$$

$$\theta_{dn}^{\mathrm{RTC}} = \mathcal{A}_{\mathrm{AST}}(d, n; \Pi_{dn}^{\mathrm{RTC}}). \tag{35}$$

For an RTC signal relation

$$y_{dn} = \sum_j L_{nj} x_{dj}, \tag{36}$$

the sky response retains L_{nj} and the full set $\{\theta_{dj}^{A}\}$ jointly. In general,

$$\text{sky-response}(L_{nj}, \{\theta_{dj}^{A}\}) \neq \sum_j L_{nj} \theta_{dj}^{A}. \tag{37}$$

When and only when an exact MAP plan Π^{MAP} requests it, a delegated materialization has the typed parent relation

$$G_{pi}^{\mathrm{materialized}} = \mathcal{M}_{\mathrm{AST}}\left(\Pi_i^{\mathrm{AST}}, \mathcal{W}_o, [\Pi_i^{\mathrm{RTC}}]_{\text{if applicable}}, \Pi_{\text{kernel/support/boundary/normalization}}^{\mathrm{MAP}}\right). \tag{38}$$

$\mathcal{M}_{\mathrm{AST}}$ denotes a materialization service, not scientific selection or ownership of G_{pi}.

5 Declared assumptions and conditional adequacy

These are bounded assumptions of the v0.1 scientific object, not achieved claims. If a required assumption is false, unsupported, or unavailable, the dependent output is unavailable or fails closed as stated by the requirements.

`SCI-AST-ASM-001` — **Authoritative upstream selection.** The exact ALIGN bundle, pointing-support record selection, physical-center selection, frame plan, RTC plan when used, and MAP projection plan when used are supplied by their named authorities. AST validates and consumes these selections but does not create them. The exact per-family center owner remains open under `AST-OWNER-Q002`.

`SCI-AST-ASM-002` — **Exact aligned field registry.** Every required boresight and current observing-state field arrives with stable identity, registry version, unit, unchanged producer frame, topology, operator, source relation, support, origin, validity, and uncertainty availability. The exact registry and composition-history semantics remain open under `AST-OWNER-Q001`; outputs needing an unresolved field are unavailable rather than inferred.

`SCI-AST-ASM-003` — **Immutable selected geometry.** A selected geometry instance is bound to one immutable APT realization and complete approved lineage. The exact producer, association authority/schema, and cross-realization transform remain open under `AST-OWNER-Q006`; an unresolved instance cannot be used unconditionally.

`SCI-AST-ASM-004` — **No hidden commutation.** The ordered composition in Section 3.5 is physical authority. Zero-offset or infinitesimal cases are limiting cases only and do not authorize reordering, double application, or removal of lineage.

`SCI-AST-ASM-005` — **Bounded small-angle production.** The established small-angle production relation is admissible only under a preregistered record naming the field/elevation/time domain, detector population, coordinate and pixel error metrics, required product use, tolerance owner and value, and fail-closed behavior. Exact spherical composition is the independent offline oracle. The owner and quantitative record remain open under `AST-OWNER-Q004`; no universal threshold or automatic fallback is created here.

`SCI-AST-ASM-006` — **Time adequacy.** A time representation is adequate only when a preregistered angular-error analysis bounds the induced astrometric error for the named use. Unused digits and an arbitrary microsecond target are not authority. The quantitative time bound remains unavailable under `AST-OWNER-Q004`.

`SCI-AST-ASM-007` — **Large-array precision adequacy.** Compact WCS authorities and fitted/catalog coordinates use adequate high precision. A representation for large coordinate arrays is admitted only after a preregistered empirical coordinate/pixel adequacy bound for the required product use. The bound remains unavailable under `AST-OWNER-Q004`.

`SCI-AST-ASM-008` — **Named frame semantics.** A requested frame transform is available only with declared source/target identities, epoch/equinox, time scale, site, model/version, input availability, and EOP/refraction applicability. Ordinary native AltAz paths do not acquire unnecessary EOP/refraction requirements, and missing equatorial authority is not repaired by relabeling.

`SCI-AST-ASM-009` — **TAN operating support.** The mathematical forward TAN domain is finite $D > 0$. A product may impose a tighter footprint-derived operating support only through a separately preregistered authority, metric, and failure rule. This package invents no positive universal $D_{\min}$.

`SCI-AST-ASM-010` — **Conditional uncertainty.** Only explicitly available map-center input covariance and cross-covariance terms are propagated. The producer, conditioning, and per-family availability of those terms remain open under `AST-OWNER-Q003`. Unavailable is not zero.

`SCI-AST-ASM-011` — **Typed Jacobian request.** A map-center Jacobian exists as an available product only after a named family requests exact input/output domains, axis order, units, and direction. Those family-specific requests remain open under `AST-OWNER-Q007`.

`SCI-AST-ASM-012` — **RTC role is explicit.** RTC-grid coordinates or response-aware use exist only for named families with an exact RTC profile instance. The requesting families and profile owner remain open under `AST-OWNER-Q008`. Missing RTC parent facts do not alter unrelated ALIGN-grid coordinate availability.

`SCI-AST-ASM-013` — **MAP-specific retained-grid policy is external.** The deferred MAP-003 retained-WCS/crop/dual-axis disposition under `AST-OWNER-Q005` is not supplied by this v0.1 authority. Any later requirement that changes AST admission or parentage requires a named amendment; until then, MAP cropping and retained-grid policy remain MAP-owned.

`SCI-AST-ASM-014` — **Nonpolarimetric scope.** Only nonpolarimetric Stokes-I coordinate use is admitted. Polarization/HWPR science interpretation, Galactic simulation, general reprojection, source fitting, calibration, photometry, beam inference, and map accumulation are outside this authority.

SCI-AST-ASM-015 — **No achieved-status inference.** Compilation, document verification, a plausible image, a downstream fit, or existing operational use cannot establish scientific approval, implementation conformity, observational performance, validation, freeze, or production readiness.

6 Edge cases and falsifiable predictions

Each prediction states a future truth condition for an implementation or representation claiming this contract. No prediction is reported as tested or satisfied here. Required-data absence is not a pass. A successor may append identifiers but shall not renumber these.

6.1 Correction, signal independence, and causes

SCI-AST-PRED-001. Zero correction and zero admitted detector displacement return the declared boresight to numerical roundoff; if that boresight is the selected TAN center, the continuous tangent coordinate is zero and the FITS coordinate is CRPIX.

SCI-AST-PRED-002. Injecting +1 arcsecond separately in each declared tangent correction component produces positive displacement along the matching basis axis with magnitude κ in the small-offset oracle.

SCI-AST-PRED-003. A declared longitude coordinate increment and an equal-valued on-sky tangent component differ by the $\cos b$ factor in Equation 11; conflating them produces a latitude/elevation-dependent error.

SCI-AST-PRED-004. For bracketed MJD support, the two endpoints and midpoint reproduce $\boldsymbol{c}_0$, $(\boldsymbol{c}_0 + \boldsymbol{c}_1)/2$, and $\boldsymbol{c}_1$, and any affine correction ramp is reproduced.

SCI-AST-PRED-005. Occurrences immediately outside either MJD endpoint, or fixtures with equal, reversed, mixed, or non-finite support, produce no clamped, nearest, extrapolated, or stale correction output.

SCI-AST-PRED-006. Explicit legacy observation-span mode reproduces its exact first and last admitted occurrence corrections and affine midpoint, records both endpoint identities, and never identifies itself as bracketed MJD.

SCI-AST-PRED-007. A one-sample legacy span is admitted only when both endpoint corrections are exactly equal; unequal endpoints make the dependent output unavailable.

SCI-AST-PRED-008. A correlated two-support covariance fixture reproduces Equation 16; a fully shared systematic does not artificially shrink at the midpoint.

SCI-AST-PRED-009. With exact occurrence/time/mapping, geometry, observing state, correction, frame, and product inputs, toggling a numerical x or r payload between finite, invalid, non-finite, unavailable, and not requested does not by itself change the AST coordinate.

SCI-AST-PRED-010. Fixtures varying each of the five cause types independently retain five independently reportable states: no signal or AST state rescues an invalid ALIGN mapping, and no coordinate fact alone authorizes downstream eligibility.

6.2 Geometry, composition, topology, and frames

SCI-AST-PRED-011. A finite non-collinear pointing correction and detector displacement match the declared ordered composition and differ from reversed composition by the expected nonzero higher-order term.

SCI-AST-PRED-012. Permuting geometry rows leaves every detector coordinate unchanged when an admitted keyed binding is used; a proven observation-local row mode is stable only inside its declared binding, while bare coincidence, duplicate keys, or missing keys fail.

SCI-AST-PRED-013. Changing any component of K_γ between pointing support and science observation makes the dependent correction/coordinate unavailable when no authorized equivalence transform is supplied.

SCI-AST-PRED-014. A supplied cross-realization transform is accepted only when both immutable parents, direction/units, domain, covariance/cross-covariance, transform uncertainty, authority, version, and failure behavior are complete; removing any required item makes the dependent transfer unavailable.

SCI-AST-PRED-015. Field rotation responds to the current aligned science-occurrence time/elevation fixture and is invariant to an unrelated elevation stored on a bracketing pointing record.

SCI-AST-PRED-016. A geometry angle or rotation-law query just outside its declared time/elevation support is unavailable and is neither clamped nor replaced by an unversioned parallactic-angle calculation.

SCI-AST-PRED-017. Fixtures with unavailable pivot covariance or unresolved affine translation, scale, skew, or shear retain those typed limitations; their outputs do not equal a silently zeroed or identity-modeled uncertainty claim.

SCI-AST-PRED-018. Directions and centers straddling $0/2\pi$ use the short signed separation, yield a continuous tangent projection, and persist normalized longitude.

SCI-AST-PRED-019. Near either coordinate pole, vector direction and in-domain round-trip remain finite, while an ambiguous longitude-only input does not establish a coordinate.

SCI-AST-PRED-020. Identical physical directions requested under Point/OOF/Beammap and Science roles produce their declared AltAz-arcsecond and equatorial-J2000- degree identities; relabeling one as the other without a named transform fails.

6.3 TAN, WCS, pixels, and product availability

SCI-AST-PRED-021. The selected TAN center maps exactly to $(0, 0)$; finite $D > 0$ is in the mathematical domain, while $D \le 0$ or non-finite D/output is invalid. No fixture is rejected solely by an invented universal positive $D_{\min}$.

SCI-AST-PRED-022. Random preregistered in-domain unit directions in both admitted frame families satisfy inverse-TAN(forward-TAN($\boldsymbol{r}$)) $= \boldsymbol{r}$ to the preregistered angular tolerance.

SCI-AST-PRED-023. The exact upstream-selected physical center, and no center inferred from bounds, source fit, coverage, or array contents, maps to tangent zero and the realized CRPIX.

SCI-AST-PRED-024. Only the exact six-field legacy all-zero pattern enters `automatic`; changing one field to a nonzero or non-finite value fails as unsupported explicit control, while numeric zero in unrelated data remains numeric.

SCI-AST-PRED-025. Removing or making ambiguous the selected center makes the affected automatic WCS unavailable without searching, fitting, recentering, or reusing a prior center.

SCI-AST-PRED-026. Positive tangent-axis perturbations move continuous pixels in the directions predicted by the realized CD-equivalent matrix; transpose, hidden flip, and swapped-axis fixtures fail.

SCI-AST-PRED-027. Odd and even geometric-center policies satisfy Equation 28, including half-integer CRPIX for even dimensions; every different declared policy still maps CRVAL exactly to its CRPIX.

SCI-AST-PRED-028. Pixel centers satisfy Equation 29; the half-open nearest-center rule includes its lower edge, excludes its upper edge, and does not depend on language-specific integer casting.

SCI-AST-PRED-029. Subpixel dithers move the continuous WCS coordinate continuously and change the nominal discrete pixel only at the declared boundary; neither event changes MAP coverage or claims coordinate precision.

SCI-AST-PRED-030. Missing, NaN, or infinite required direction, tangent, WCS, or continuous-pixel inputs produce explicit invalidity/unavailability and never a cast integer, edge pixel, center, or stale coordinate.

SCI-AST-PRED-031. A full-TOD request may retain coordinate arrays with compact interpretation state, whereas an otherwise equivalent mini-TOD request reports those arrays unavailable rather than silently omitting their identity.

6.4 Jacobian, uncertainty, RTC, and MAP

SCI-AST-PRED-032. For every claimed available family/domain pair, the typed map-center Jacobian agrees with a preregistered symmetric finite-difference or validated automatic/analytic derivative oracle, including unit, axis-sign, center, pixel, and near-wrap fixtures.

SCI-AST-PRED-033. Small correlated map-center input draws reproduce Equation 33 in the declared linear domain, including the supplied cross term; removing a term changes its state to unavailable rather than numeric zero.

SCI-AST-PRED-034. Without a named family and exact input/output domains, the `astrometric_map_center_jacobian` is unavailable and is not inferred from shape, WCS, or consumer name.

SCI-AST-PRED-035. An ordinary phase-zero RTC fixture may make $\theta_{dn}^{\mathrm{RTC}}$ numerically equal to one θ_{dj}^{A}, but the two outputs retain different exact parent identities and roles.

SCI-AST-PRED-036. Removing any required RTC plan, grid, occurrence, time, phase/delay, segment, decimation, support, response, or correction fact makes only the dependent RTC-grid coordinate, response composition, or delegated G_{pi} unavailable; unrelated ALIGN-grid coordinates remain unchanged.

SCI-AST-PRED-037. For a temporal operator with negative coefficients or coordinates crossing circular topology, direct $\sum_j L_{nj}\theta_{dj}^{A}$ differs from the sky response obtained by retaining the operator and its full coordinate support, falsifying angular signal filtering.

SCI-AST-PRED-038. Changing the RTC output-grid parent or MAP projection-plan parent while holding numerical coordinates and array shape fixed changes the identity of every delegated G_{pi} artifact; similar shape is not compatibility.

SCI-AST-PRED-039. AST refuses delegated G_{pi} materialization when the MAP kernel, support, normalization, boundary, conservation, artifact form, plan/version, or response/covariance role is missing or ambiguous.

SCI-AST-PRED-040. Two MAP-owned deposition plans using the same AST continuous coordinates may realize different valid G_{pi} operators without changing the underlying AST coordinate/WCS identity, demonstrating the ownership boundary.

6.5 Lifecycle, parity, failure, and empirical gates

SCI-AST-PRED-041. A valid–missing–valid observation sequence with different correction, geometry, center, WCS, and RTC modes shows no inherited prior state; recovery after failure constructs a clean candidate.

SCI-AST-PRED-042. Permuted scheduling and thread counts preserve per-parent coordinates, pixels, validity, counts, serialized WCS, and deterministic provenance order under the declared numerical policy.

SCI-AST-PRED-043. Identical supported abstract inputs through real and simulation paths produce the same scientific coordinate and validity; only declared origin/generator/version/seed provenance differs.

SCI-AST-PRED-044. Injected required coordinate, WCS, provenance, and product-write failures each produce unsuccessful required-product/reduction state and no apparently complete mislabeled artifact.

SCI-AST-PRED-045. Within a preregistered small-angle domain, the production operator's coordinate and pixel errors relative to the exact spherical oracle satisfy the named tolerance; outside the domain or without the record, the dependent adequacy claim/output fails closed rather than switching algorithms silently.

SCI-AST-PRED-046. Preregistered time-quantization and large-array representation perturbations satisfy their named angular/pixel error bounds for the required use; a missing owner, metric, domain, tolerance, or failure rule cannot yield an achieved-adequacy claim.

SCI-AST-PRED-047. Unsupported Galactic/other-frame simulation or polarization-science requests fail at admission and do not emerge as relabeled supported coordinates.

SCI-AST-PRED-048. A downstream source, pointing, OOF, or Beammap fit may succeed or fail without changing the already realized AST coordinate identity or retroactively establishing AST validity.

SCI-AST-PRED-049. For every persisted coordinate, compact provenance reconstructs the requested, effective, observation-resolved, and realized policy; changing any parent or version changes the reconstructed identity rather than silently aliasing it.

SCI-AST-PRED-050. A compilation, clean render, or plausible coordinate image alone does not change the explicit states of scientific approval, implementation conformity, empirical validation, freeze, or production readiness.

Conformance disposition

Every requirement remains prospective until separately authorized evidence binds it to an exact implementation and representation. Open owner questions are not implementation choices: they keep only their dependent outputs or quantitative claims unavailable. Passing a subset of predictions cannot establish scientific approval, overall conformity, validation, freeze, or production readiness.